

benchmarking Android systems, such as tablets and automotive SoC.

Native Development Kit

Android supports a Native Development Kit to embed components that make use of native C/C++ code in Android applications. The NDK helps address the needs of performance and graphics-sensitive applications.

Optimised graphics and media support

Android provides support for a wide range of media formats through StageFright, its custom media framework. It also provides its own 2D graphics library, but relies on OpenGL ES for its 3D capabilities. This feature makes it feasible to create small-sized embedded systems with high-end audio and video capabilities.

Telephony support

Android supports telephony, which is dependent on hardware. For this feature to work, device manufacturers need to create a HAL module to interface with their hardware, which is integrated as part of the Android build system.

Wireless connectivity

Android supports most wireless connectivity options, like Bluetooth, EDGE, 3G and Wi-Fi. This enables the embedded device to have a wide range of connectivity options to any third-party systems.

Licensing

Android licensing terms are pretty friendly for both commercial and free open source applications. This is because all its core packages are open sourced under the terms of the Apache 2.0 licence.

Customisability

Android has well-defined interfacing between the framework and its components. This helps to enhance or replace components as per the desired functionality. For example, the default launcher application can be enhanced or replaced with a different launcher application code base. The platform can be enhanced to support additional features

and hardware, as desired.

The Android advantage

So let's understand how Android can help give an extra edge to your embedded system in the market.

A well accepted and recognised UX

Android user experience (UX) designs such as the Frog's Feel UX, HTC Sense and many others have raised UX standards to quite a high level. This means that your system can enjoy the high-end UX capabilities that Android supports. With this edge, the learning curve for end users of your embedded system remains no longer a big concern.

Forward-compatible apps

All APIs provided in the application framework are meant to be forward-compatible. Hence, apps developed for one version would continue working in future Android versions. However, platform-level changes would require to be ported when planning to upgrade the Android version of your custom solution.

Wide choice of hardware configuration

Android has wide support from the OEM and SoC community. Though primarily supporting ARM-based SoCs, Android now also supports x86-based SoCs. This provides for a wide range of hardware configurations to choose from, depending on what fits your budget and system requirements.

Hassle-free supply chain commoditisation

No doubt—Android, with its solid foundation on Linux, sound ecosystem and developer community, together with the cutting edge advantages, has truly proven itself as the platform of choice for any modern embedded system. 

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